



FINAL YEAR PROJECT (FYP)
FRAMEWORK OF FACULTY OF ENGINEERING
2024

Acknowledgement

With reference to the 41st BoS meeting and further on the recommendations/ directives of Chairman ECE/ Acting Dean FoE, the following committees was constituted to prepare a comprehensive document for implementation of standard operation procedures for Final Year Project (FYP) framework.

- Dr. Bilal Ijaz, Convener- ECE Islamabad Campus
- Dr. Mirza Tariq Humayun, Member- ECE Lahore Campus
- Dr. Muhammad Kamran, Member- ECE Wah Campus
- Dr. Aamir Shahzad, Member-ECE Abbottabad Campus
- Dr. Babar Sattar Khan, Member- ECE Attock Campus
- Dr. Atif Mehmood- Member ECE Sahiwal Campus

Table of Contents

1. Introduction	4
2. Objective	4
3. Eligibility Criteria	4
4. Procedures	4
5. Risks and Mitigation	6
6. SOP Review Period	6
7. Flow Chart of FYP procedures	7
8. Course Learning Outcomes	7
9. Anti-Plagiarism Policy	8
10. Annexure I	9
11. Annexure II	10
12. Annexure III	11
13. Annexure IV	12
14. Annexure V	13
15. Annexure VI	14
16. Final Year Project Rubrics Spring 2024 Semester	15
17. Oral Presentation Rubric – 8th Semester	18
18. Written Report Rubric – 8th Semester	19
19. Sample Project Report	20

1. Introduction:

The Final Year Project is the largest piece of work assigned to students. It is intended both to consolidate the skills gained throughout the degree program as well as prepare students to undertake and participate in projects upon graduation. Thus, the aims as well as the criteria of assessment of Final Year Project are rather wide-ranging.

2. Objective:

The objectives of the FYP are to enable the student to integrate and apply the knowledge and skills acquired in various courses. Furthermore, the student should be able to independently acquire requisite knowledge to propose a technical solution to a complex engineering problem in a timely and ethical manner. Finally, the student should be able to work independently or in a team and be able to communicate and justify the proposed solution effectively.

The student should be able to demonstrate a broad understanding of the UN SDG global challenges. In addition, the technical solution presented and accomplished in FYP should be mapped to the relevant UN sustainable development goals (SDG) 2015 where applicable.

3. Eligibility Criteria:

As per CUI policy, the students are eligible to register FYP Part-I at the start of their 7th semester provided they have completed a minimum 80 credit hours. As the right selection of project and finding the supervisor has always been an important step and is time taking, therefore the departments within the Faculty of Engineering (FoE) offers FYP at the beginning of the 6th semester to ensure that students can effectively work on their FYP right after the commencement of the 7th semester.

4. Procedures:

An FYP offered by the Departments within the FoE is composed of the following important steps:

- a) FYP offering by faculty members:** In the beginning of semester, faculty members are requested to propose the projects by submitting the brief project description to Project Management Committee (PMC). The lists of offered projects are displayed in the department as well as provided to all students through undergraduate office. Students are supposed to select the project out of the list and discuss it with the relevant faculty member/ project supervisor.
- b) Project enrollment within department in 6th Semester:** With the consent of the supervisor, the student submits the kickoff report (KOR) along with Project Accept / Reject form to the PMC office (Sample KOR and Project Accept / Reject Form is attached in annexure-I & II).
- c) FYP Part-1 Registration (03 Credit Hours) 7th Semester:** On satisfactory performance and recommendation of FYP supervisor the students register 03 credit hours on the portal (Part-1 of FYP) in start of 7th semester.

- d) Evaluation of Part-1 (end of 7th semester):** In this phase, each project is evaluated by the supervisor along with the two faculty members as examiners at the end of 7th semester. This panel of three examiners observes performance based on the defined rubrics. Based on evaluation of internal examiners and project supervisor, PMC compiles the result of Final Year Project (Part 1) and forwards it to the Examination Department (Sample evaluation forms for 7th semester FYP are attached in annexure-III).
- e) FYP Part-2 Registration (03 Credit Hours) 8th Semester:** On successful completion of Part-1, the students are allowed to register for the remaining 03 credit hours in the 8th semester. If a student fails to earn satisfactory status in the seventh semester, then he/she is required to register again 03 credit hours of seventh semester.
- f) Submission of First Draft of Thesis:** The students are required to submit the first draft of their FYP report before midterm. Each student submits three hard copies of the first draft to PMC. The PMC forwards one copy to supervisor and one to each of internal examiners for their comments and suggestions.
- g) Evaluation of FYP Part-2 (8th Semester):** The part -2 evaluation is carried out in the following two steps.
- i. **Internal Evaluation:** The students who have submitted the first draft of their work are allowed to present their work at this stage. The PMC office makes the first presentation schedule and invites the supervisors along with internal faculty members to evaluate the work so far. The panel of three examiners thoroughly review the work and give comments for consideration before the final defense.
 - ii. **External Evaluation:** In the final evaluation FYP is evaluated by a team comprising supervisor, two internal examiners and one external examiner. External examiners are invited from the reputed university or organization approved from the Academic Council. The panel evaluate the performance based on approved rubrics (Sample rubrics is attached in annexure IV)
- h) Results compilation and Submission of final thesis copies to exam department:** The PMC office compiles the results (Sample result forms are attached in the annexure V) finally; the students submit five copies of final draft of thesis with the modifications suggested by the internal and external examiners. (Thesis format sample is attached in annexure-VI).

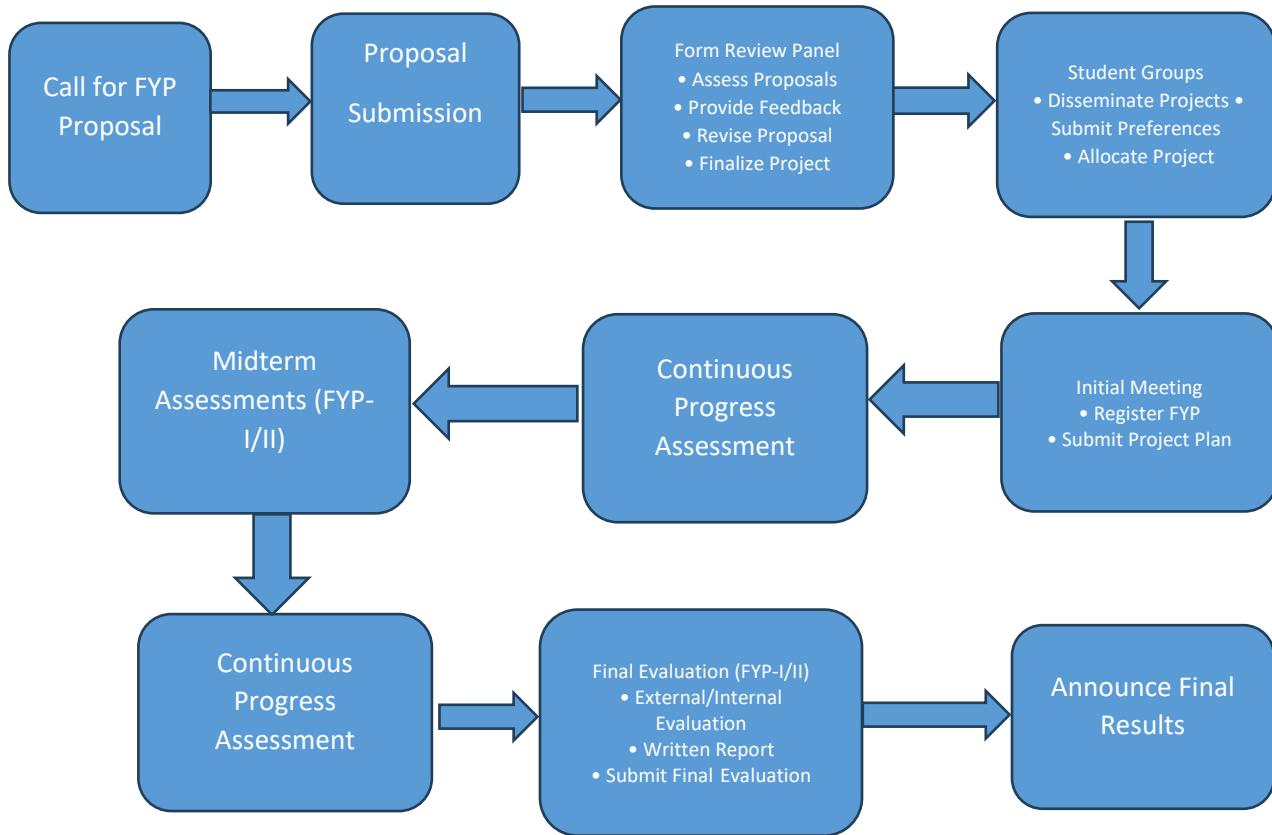
5. Risks and Mitigation

Risks	Mitigation
Insufficient number of projects proposed	Head of Department ensures each faculty submits a minimum number of potential project proposals. Students are also encouraged to propose their own ideas.
Student does not register by the deadline	Student is permitted to register late subject to Head of Department approval.
Assigned supervisor is not available for long duration	FYP committee arranges the replacement for the supervisor.
Project is not completed by the deadline	Supervisor ensures students work in project lab for one credit hour in FYP-I and five credit hours in FYP-II. Regular evaluation is conducted by the supervisor during the project.
Late submission of report	Late submission is penalized by deducting Marks.

6. SOP Review Period

In order to implement the mechanism of Continuous Quality Improvement (CQI), the standard operation procedures for the FYP will be reviewed annually or need basis (as and when required).

7. Flow Chart of FYP procedures



8. Course Learning Outcomes (CLOs)

The aim of the FYP is to enable the student to integrate and apply the knowledge and skills acquired in various courses. Furthermore, the student should be able to independently acquire requisite knowledge to propose a technical solution to a complex engineering problem in a timely and ethical manner. Finally, the student should be able to work independently or in a team and be able to communicate and justify the proposed solution effectively.

After successfully completing FYP, the students will be able to:

1. Utilize fundamental knowledge of science and engineering to identify a complex engineering problem and propose a viable solution. **(C3-PLO1)**
2. Provide a critical review of previous work in the field, and a theoretical foundation for the approach taken in the project. **(C4-PLO2)**
3. Design a system, component, or process that provides a feasible solution to a complex engineering problem which subsequently leads in achieving UN sustainable development goals. **(C5-PLO3)**
4. Investigate the solution in a methodical way including but not limited to necessary

literature survey, experiment design and synthesis of information. **(C6-PLO4)**

5. Describe the proposed solution's societal implications such as its health and safety aspects in accordance with UN sustainable development goals. **(A3-PLO6)**
6. Describe the proposed solution's sustainable development considerations keeping in view the criteria given in UN sustainable development goals. **(A3-PLO7)**
7. Develop a solution in accordance with the norms of engineering practice and professional ethics also proposed in UN sustainable development goal targeting responsible consumption and production. **(A3- PLO8)**
8. Contribute to the project goals both as an individual and a team player. **(A3-PLO9)**
9. Effectively communicate and justify the developed solution and be able to demonstrate the project to any general audience. **(A2-PLO10)**
10. Demonstrate management skills to achieve project goals in a timely manner. **(A3-PLO11)**
11. Demonstrate an ability to independently seek and acquire knowledge required for successful project completion. **(A3-PLO12)**

PLOs	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO 9	PLO10	PLO11	PLO12
CLOs												
CLO1	C3											
CLO2		C4										
CLO3			C5									
CLO4				C6								
CLO5						A3						
CLO6							A3					
CLO7								A3				
CLO8									A3			
CLO9										A2		
CLO10											A3	
CLO11												A3

9. Anti-Plagiarism Policy

The Academic Council in its 38th meeting adopted the Higher Education Commission (HEC)'s Anti-plagiarism Policy – Version 2.0 under notification No. CUI-Reg/Notif-158/23/944. The same policy is applicable on all final FYP reports.

The annexures attached are taken from the Department of Electrical and Computer Engineering FYP handbook for reference only. Other departments within the FoE may replace them with their own annexures.



COMSATS University, Islamabad Campus
(Department of Electrical and Computer Engineering)
Kickoff Report

Annexure-I

Project Title	Submitted Date

Group Members:

S.No	Name	Registration No.	Mobile No.	Email	Signature
1	As written on Student Card	CIIT/FA20-BEE-001/ISB as written on Student Card			
2					
3					
4					

Supervisor(s) Detail:

Role	Name	Designation	Mobile No.	Email	Signature
Internal Supervisor					
Co- Supervisor (if any)					

Instructions:

- Write the name and registration number which is on the student ID card.
- **Handwritten form will not be accepted.**
- **All the supervisors have the soft copy of this form.**
- **Attach Project Accept / Reject form.**
- Kindly send the soft copy of unsigned version on Email: **mramzan_malik@comsats.edu.pk**



COMSATS University, Islamabad Campus
(Department of Electrical and Computer Engineering)
Accept and Reject form

Annexure-II

Project Title:

Industrial Collaboration

Yes

No

Name of Funding Organization:

Area of Specialization:

Antenna & Microwave

☐

Communication

☐

Networks

☐

Signal Processing

☐

Controls/Digital Systems

☐

Computer Vision

☐

Integrated Circuits & Systems

☐

Cybernetic Robotics

☐

Renewable Energy

☐

Artificial Intelligence/ML

☐

Internet of Things

☐

& Power Engineering

Abstract: (Please use additional page if needed)

(Supervisor Name & Signature)

Email

Cell No.

UNSDGs:

S. No	Goal	Addressed	S. No	Goal	Addressed	S. No	Goal	Addressed
1	No Poverty		7	Affordable and Clean Energy		13	Climate Action	
2	Zero Hunger		8	Decent Work and Economic Growth		14	Life Below Water	
3	Good Health and Well-Being		9	Industry, Innovation and Infrastructure		15	Life on land	
4	Quality Education		10	Reduced Inequalities		16	Peace, Justice and Strong Institutions	
5	Gender Education		11	Sustainable Cities and Communities		17	Partnership for the Goal	
6	Clean Water and Sanitation		12	Responsible Consumption and Production				

Note for supervisor: Kindly ensure at least one UN SDG is addressed in the FYP

(Name & Signature of Supervisor)



COMSATS University Islamabad, Campus
(Department of Electrical and Computer Engineering)
Evaluation of Final Year Project Part-I (Spring-2024)
(To be filled by Internal Evaluators)

Annexure-III

Semester: 7th

Date: _____

Project Title: _____

Name: _____

Registration Number: _____

Evaluation of Project Design Rubrics

R1 (Identification of the problem)	R2 (Project analysis)	R3 (Design and Development)	R4 (Investigation)

Evaluation of Oral Presentation Rubrics

OR1 (Content)	OR2 (Presentation Skills)	OR3 (Organization)	OR4 (Handling of Questions)

Name of Evaluator: _____

Signature of Evaluator: _____

Maximum Marks of each Rubric: 100



COMSATS University, Islamabad Campus
(Department of Electrical and Computer Engineering)
Evaluation of Final Year Project Part-I (Spring-2024)
(To be filled by Supervisor)

Annexure-IV

Semester: 7th

Date: _____

Name: _____

Registration Number: _____

Evaluation of Project Design Rubrics

R1 (Identification of the problem)	R2 (Project analysis)	R3 (Design and Development)	R4 (Investigation)	R6 (The Engineer and Society)
R7 (Environment and Sustainability)	R8 (Ethical Practice)	R9 (Individual and Teamwork)	R10 (Project Management)	

Evaluation of Oral Presentation Rubrics

OR1 (Content)	OR2 (Presentation Skills)	OR3 (Organization)	OR4 (Handling of Questions)

Evaluation of Written Rubrics

WR1 (Format and Organization)	WR2 (Language and Grammar)	WR3 (Equations, Numerical Usage, and Illustrations)

Name of Supervisor: _____

Signature of Supervisor: _____

Maximum Marks of each Rubric: 100



COMSATS University, Islamabad Campus
(Department of Electrical and Computer Engineering)
Evaluation of Final Year Project Part-II (Spring-2024)
(To be filled by Internal / External Evaluators)

Annexure-V

Semester: 8th

Date: _____

Project Title: _____

Name: _____

Registration Number: _____

Evaluation of Project Design Rubrics

R2 (Project analysis)	R3 (Design and Development)	R4 (Investigation)	R5 (Knowledge Beyond Course Work)

Evaluation of Oral Presentation Rubrics

OR1 (Content)	OR2 (Presentation Skills)	OR3 (Organization)	OR4 (Handling of Questions)

Please tick the appropriate option:

Evaluator-1

☐

Evaluator-II

☐

External Evaluator

☐

Name of Evaluator: _____

Signature of Evaluator: _____

Maximum Marks of each Rubric: 100



COMSATS University, Islamabad Campus
(Department of Electrical and Computer Engineering)
Evaluation of Final Year Project Part-II (Spring-2024)
(To be filled by Supervisor)

Annexure-VI

Semester: 8th

Date: _____

Project Title: _____

Name: _____

Registration Number: _____

Evaluation of Project Design Rubrics

R2 (Project analysis)	R3 (Design and Development)	R4 (Investigation)	R5 (Knowledge Beyond Course Work)	R6 (The Engineer and Society)
R7 (Environment and Sustainability)	R8 (Ethical Practice)	R9 (Individual and Teamwork)	R10 (Project Management)	

Evaluation of Oral Presentation Rubrics

OR1 (Content)	OR2 (Presentation Skills)	OR3 (Organization)	OR4 (Handling of Questions)

Evaluation of Written Rubrics

WR1 (Format and Organization)	WR2 (Language and Grammar)	WR3 (Equations, Numerical Usage, and Illustrations)

Name of Supervisor: _____

Signature of Supervisor: _____

Maximum Marks of each Rubric: 100



Final Year Project Rubrics Spring 2024 Semester

	Excellent (85 – 100)	Good (70 - 84)	Adequate (55 - 69)	Inadequate (0 – 54)
R1: [C3-PLO1] Identification of the problem	The problem has been shown (not just stated) to exist with supporting factual evidence.	A problem statement has been stated.	The problem statement has weak support.	Problem has not been stated clearly and lacks any supporting evidence.
R2: [C4-PLO2] Project analysis	- Existing solutions to the problem with their relative merits have been discussed. - Clear expectations of project deliverables are stated with measurable performance requirements.	- Existing solutions have been stated. Additional discussion may be warranted in places. - Expectations have been stated. - Some objectives may not be measurable.	- A complete review of existing solutions and research related to this problem is not presented. - Expectations have been stated. - Most objectives are not measurable.	Connection between references and what is written is not clear. Little investigation has been done. - Expectations are not clear. - Expectations are not measurable.
R3: [C5-PLO3] Design and Development [UN SDG No 9]	- A comprehensive listing of critical design criteria is provided - All design criteria have been associated with requirement	- A comprehensive listing of critical design criteria is provided - Majority of design requirements have been identified	- The comprehensive listing of critical design criteria is incomplete or inadequate. - Some requirements have been identified	- The comprehensive listing of critical design criteria is incomplete or inadequate. - Design requirements have not been identified
R4: [C6-PLO4] Investigation	- All sub-systems and hardware and software interfaces were well-designed to extract necessary technical data. - The data were analyzed, and all valid conclusions were drawn.	- Most sub-systems and hardware and software interfaces were designed to extract necessary technical data. - The data were analyzed and most of the relevant conclusions were drawn.	- Some sub-systems and hardware and software interfaces were designed to extract technical data. - The data were analyzed, and some conclusions were drawn.	- Necessary technical data was not extracted due to poor experiment design. - The data were not properly analyzed, and incorrect conclusions were drawn.



COMSATS University Islamabad, Islamabad Campus
(Department of Electrical and Computer Engineering)

R5: [A3-PLO12] Knowledge Beyond Course Work	• A significant portion of this project involves technical information outside the scope of the undergraduate curriculum.	• Several technical aspects were new to the students and required research.	• This project contains some research but mostly involves technical information taught at the junior and senior levels.	• This project did not challenge the students to perform much research, as it relied mainly on information taught within the curriculum.
<u>Additional Assessment (for supervisor only)</u>				
R-6: [A3-PLO6] The Engineer and Society [UN SDG No 3]	• The project thoroughly addresses the impact of proposed solution on the society.	• The implications of this project have been discussed for various societal issues.	• The project briefly mentions the issues related with health and safety.	• The proposed solution lacks discussion on the societal issues.
R-7: [A3-PLO7] Environment and Sustainability [UN SDG No 11]	• The project presents a comprehensive discussion on the solution's sustainable development aspects.	• Some sustainability aspects have been discussed in this project.	• The description of the sustainability perspective is briefly mentioned in the project.	• The project lacks discussion on the sustainability aspects.



COMSATS University Islamabad, Islamabad Campus
(Department of Electrical and Computer Engineering)

R8: [A3-PLO8] Ethical Practice [UN SDG No 12]	• Demonstrates a comprehensive understanding of ethical concepts. • Can provide a comprehensive explanation of own ethical behavior, which is supported by gathering relevant facts and information.	• Demonstrates a considerable understanding of ethical concepts. • Can provide logical explanation of own ethical behavior supported by some evidence of gathering pertinent facts and information.	• Demonstrates a general understanding of ethical concepts. • Can provide a general explanation of own ethical behavior but is not sure of what supporting evidence must be gathered.	• Demonstrates superficial understanding of ethical concepts. • Can only provide a superficial explanation of own ethical behavior.
R9: [A3-PLO9] Individual and Teamwork	• Student demonstrated an effective individual technical contribution towards project completion. • Student helped fellow team members in discussing & implementing alternative solutions to their tasks.	• Student demonstrated an effective individual technical contribution towards project completion. • He/she helped fellow team members only by suggesting alternative solutions to them. • Values some peer criticism.	• Student demonstrated some individual contribution towards project completion. • Student was unable to demonstrate the capacity to help fellow team members. • Not readily willing to value constructive peer criticism.	• Student was unable to demonstrate significant individual contribution towards project completion. • Student demonstrates behavior contrary to team-spirit.
R10: [A3-PLO11] Project Management	• Well-defined project-plan listing timelines & resources • Feedback was regularly sought & incorporated • All Project deliverables timely completed & well-documented, within resources	• Well-defined project plan listing timelines & resources • Some deliverables required extra time or resources	• Project planning has some deficiencies in stated timelines and resource • Some sub-tasks partially completed, e.g., restricted functionality of hardware/software	• Lack of planning is evident • Key deliverables could not be completed • project review meetings were few and far between



COMSATS University Islamabad Campus
(Department of Electrical and Computer Engineering)

Oral Presentation Rubric – 8th Semester
[A2-PLO10]

	Excellent (85 – 100)	Good (70 - 84)	Adequate (55 - 69)	Inadequate (0 – 54)
OR1: Content	- Addresses all specified content areas. - Material abundantly supports the topic. - Use of engineering terms and jargon matches audience knowledge level.	- Addresses most content areas. - Material sufficiently supports the topic. - Use of engineering terms and jargon mostly matches audience knowledge level.	- Addresses some of the content areas. - Material minimally supports the topic. - Use of engineering terms and jargon minimally matches audience knowledge level.	- Addresses few of the content areas. - Material does not support the topic. - Use of engineering terms and jargon does not match audience knowledge level.
OR2: Presentation Skills	- Clearly heard and polished. - Attitude indicates confidence and enthusiasm. - Audience attention is constantly maintained.	- Clearly heard but not polished. - Attitude indicates confidence but not enthusiasm. - Audience attention is mostly maintained.	- Difficult to hear and/or moments of awkwardness. - Attitude indicates some lack of confidence and/or disinterest in subject. - Audience attention is minimally maintained.	- Inaudible; several awkward pauses. - Attitude indicates lack of confidence and/or disinterest in subject. - Audience attention is not maintained.
OR3: Organization	Information presented in logical and interesting sequence that the audience can easily follow.	Information presented in a logical sequence that the audience can easily follow.	Information not always presented in a logical sequence; audience has difficulty following presentation.	Information not presented a logical sequence; audience cannot understand presentation
OR4: Handling of Questions	Demonstrates full knowledge of the material; can explain and elaborate on expected questions.	Demonstrates sufficient knowledge of the material to answer expected questions.	Demonstrates difficulty answering expected questions beyond a rudimentary level.	Demonstrates an inability to answer expected questions.



COMSATS University Islamabad Campus
(Department of Electrical and Computer Engineering)

Written Report Rubric – 8th Semester
[A2-PLO10]

	Excellent (85 – 100)	Good (70 - 84)	Adequate (55 - 69)	Inadequate (0 – 54)
WR1: Format and Organization	The document is well organized and formatted aesthetically. There is a logical flow of information across chapters, sections, and paragraphs.	Mostly, the document is organized and formatted. Chapters and sections are well organized but logical flow at paragraph level could be improved.	- The document contains formatting errors. - Within sections, the order in which ideas are presented is occasionally confusing.	- The document is poorly organized and formatted. - There is no logical ordering of chapters and sections.
WR2: Language And Grammar	- Sentences are properly constructed and grammatically correct. - Engineering terminology is appropriately used.	- For the most part, sentences are complete and grammatically correct. - Engineering terminology is appropriately used on most occasions.	- Some errors in sentence structure and grammar are present. - Occasionally, Engineering terminology is used without definition.	- Poor sentence construction and use of grammar. - There is unnecessary Repetition of ideas. - Engineering terminology is minimally used.
WR3: Equations, Numerical Usage, and Illustrations, References	Equations, figures and tables are appropriately numbered and complement the text. Prior work is acknowledged through appropriate and up-to-date referencing.	- For the most part, equations, figures and tables are consistent with the text. - Prior work is mostly acknowledged through appropriate and up-to-date referencing.	Some, equations, figures and tables are inconsistent with the text. Prior work is acknowledged but some relevant references are missing.	- Relevant Equations, figures and tables are either missing or are not discussed in the text.. - Prior work is not acknowledged or quality of referencing is poor.

Project Title



Final Year Project Report

Presented

by

Student-1 Name

CIIT/FA19-BEE-051/ISB

Student-2 Name

CIIT/ FA19-BEE -050/ISB

Student-3 Name

CIIT/ FA19-BEE -049/ISB

In Partial Fulfillment

of the Requirement for the Degree of

*Bachelor of Science in Electrical (Electronics/Computer)
Engineering*

DEPARTMENT OF ELECTRICAL AND COMPUTER ENGINEERING

COMSATS UNIVERSITY ISLAMABAD

July 2024

Declaration

We, hereby declare that this project neither as a whole nor as a part there of has been copied out from any source. It is further declared that we have developed this project and the accompanied report entirely on the basis of our personal efforts made under the sincere guidance of our supervisor. No portion of the work presented in this report has been submitted in the support of any other degree or qualification of this or any other University or Institute of learning, if found we shall stand responsible.

Signature:_____

Name

Signature:_____

Name

Signature:_____

Name

COMSATS UNIVERSITY ISLAMABAD

July 2024

Write Project Title

An Undergraduate Final Year Project Report submitted to the

Department of

ELECTRICAL AND COMPUTER ENGINEERING

As a Partial Fulfillment for the award of Degree

*Bachelor of Science in Electrical (Computer/Electronics)
Engineering*

by

Name

Registration Number

Student-1

CIIT/FA20-BEE-001/ISB

Student-2

CIIT/FA20-BEE-001/ISB

Student-3

CIIT/FA20-BEE-001 /ISB

Supervised by

Name

Designation,

Department of Electrical and Computer Engineering

CU Islamabad

COMSATS UNIVERSITY ISLAMABAD

July 2024

Final Approval

This Project Titled
Write Project Title
Submitted for the Degree of
Bachelors of Science in Electrical (Computer/Electronics)
Engineering

by

Name	Registration Number
Student-1	CIIT/FA19-BEE-001/ISB
Student-2	CIIT/ FA19-BEE-008/ISB
Student-3	CIIT/ FA19-BEE-010/ISB

has been approved for
COMSATS UNIVERSITY ISLAMABAD

Supervisor
Name,
Designation

Internal Examiner-1
Name,
Designation

Internal Examiner-2
Name,
Designation

External Examiner
Name,
Designation

Head
Department of Electrical and Computer Engineering

Dedication

Body text the font must be as follows (watch the word font bar at the top). Text Font is 12 (Times New Roman), justified

Acknowledgements

Body text the font must be as follows (watch the word font bar at the top). Text Font is 12 (Times New Roman), justified

Name Student-1

Name Student-2

Name Student-3

Table of Contents

1.	Introduction	1
1.1	Purpose.....	2
1.2	Document Convention.....	2
1.2.1	Definitions.....	2
1.3	Intended Audience and Reading Suggestions.....	3
1.4	Project Scope	3
	Description.....	4
1.4.1	Product perspective.....	4
1.4.2	Product Features.....	4
1.4.3	Operating Environment.....	5
1.5	Assumptions and Dependencies.....	5
1.5.1	Assumptions.....	5
1.5.2	Dependencies.....	5
1.6	Broader Impact (UN SDGs).....	5
1.6.1	Targeted SDGs.....	5
1.6.2	Potential Mapping (checklist).....	5
2.	Literature Review	6
2.1	System Analysis.....	6
2.2	Software Features.....	6
3.	Modeling and Simulation	7
4.	Hardware Implementation, Testing and Results	8
5.	Discussion	9
6.	Future Work	10
Appendix-A		60
Appendix-B		70
Appendix-C		79
Bibliography		79

List of Acronyms

ROC.....Region of Convergence

FAR.....False Acceptance Rate

FRR.....False Rejection Rate

ERR.....Equal Error Rate

List of Figures

1.1	ROC curves for two out of 40 signers.....	1
1.2	Caption.....	pg#
1.3	Caption.....	pg#
1.4	Caption.....	pg#
1.5	Caption.....	pg#
2.1	Caption.....	pg#
2.2	Caption.....	pg#
2.3	Caption.....	pg#
2.4	Caption.....	pg#
2.5	Caption.....	pg#

List of Tables

1.1	Equal Error Rates (EER) For 14 Signers.....	1
1.2	Caption.....	pg#
1.3	Caption.....	pg#
1.4	Caption.....	pg#
1.5	Caption.....	pg#
2.1	Caption.....	pg#
2.2	Caption.....	pg#
2.3	Caption.....	pg#
2.4	Caption.....	pg#
2.5	Caption.....	pg#

Abstract

Body text the font must be as follows (watch the word font bar at the top). Text Font is 12 (Times New Roman), justified. List of acronyms, list of figures and list of tables must be mentioned in the table of contents.